

Mary Appah

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SUMMARY

Dynamic Biostatistician with over 4 years of experience in developing advanced statistical models and methodologies to enhance clinical trial design and data analysis. Expertise in Bayesian statistical modeling, adaptive randomization, and mediation analysis. Proven track record of optimizing decision-making and improving research reproducibility. Committed to leveraging robust analytical skills and innovative approaches to contribute meaningfully to research objectives in biostatistics and public health.

WORK EXPERIENCE

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| 08/2020 – Present | Research Associate (PhD research), UAB, School of Public Health Birmingham <ul style="list-style-type: none">• Designed and implemented advanced statistical techniques to mitigate the impact of outliers on mediation effect estimation in clinical trial data.• Enhanced statistical methodologies to improve efficiency, decision-making, and reliability in adaptive clinical trial designs for drug development.• Applied structural equation modeling (SEM) and causal inference methods to assess the direct and indirect effects of treatments on clinical outcomes.• Utilized quantile regression, LASSO, Ridge, and Elastic Net to improve estimation accuracy and handle multicollinearity in high-dimensional clinical trial data.• Incorporated Bayesian methods to improve inference, enable adaptive decision-making, and integrate prior knowledge into clinical trial analysis.• Implemented dynamic randomization techniques to optimize patient allocation and improve the efficiency of treatment evaluation.• Executed large-scale simulations to assess the robustness of statistical approaches under various clinical trial scenarios.• Developed rigorous analytical workflows that enhance statistical efficiency and reproducibility in clinical research. |
| 09/2018 – 03/2020 | Data Analyst, Heersink School of Medicine, Department of Neurology, UAB Birmingham, USA <ul style="list-style-type: none">• Established computational approaches and analytical tools to analyze microbiome data in patients with Parkinson's disease.• Evaluated large molecular datasets including genomic sequence data from patient samples for clinical research purposes.• Devised bioinformatics algorithms including unsupervised and supervised machine learning to cluster and classify gut microorganisms.• Created clinical data reports, ensuring accuracy for seamless integration into data repositories.• Contributed to the development of standardized protocols for sample collection and processing.• Assisted in the development of data management systems and repositories.• Employed R and SAS to develop robust computational methods, analyzing large genomic sequences and microbiome data sets. |

08/2022 – 05/2026	Instructor, Statistics, UAB Department of Biostatistics	Birmingham, USA	• Developed comprehensive teaching materials and curriculum centered on the use of data analytics in biostatistics. • Formulated statistical models to help students understand data-driven decision-making and problem-solving within healthcare and life sciences. • Mentored students on analytical tools such as R and Excel for effective data analysis. • Promoted the application of technology solutions in addressing real-world data problems, fostering teamwork and interpersonal skills through group assignments and discussions. • Implemented innovative teaching methodologies to facilitate active learning and enhance student engagement.
08/2020 – 05/2026	Teaching Assistant, UAB Department of Biostatistics	Birmingham, USA	• Supervised and assisted in the teaching of the statistics course for first-year undergraduate students. • Provided extra-classroom advice and assistance to students, promoting academic success. • Led small group discussions, review sessions, and problem-solving activities to reinforce course concepts. • Graded assignments, quizzes, and exams while offering constructive feedback to students. • Assisted in developing course materials and maintaining consistency in course delivery. • Participated in professional development activities to enhance teaching skills. • Held regular office hours and utilized instructional technologies to support student learning.
08/2016 – 05/2018	Teaching Assistant, Northern Arizona University	Arizona, USA	• Supervised and assisted in the teaching of the statistics course for first-year undergraduate students. • Provided extra-classroom advice and assistance to students, enhancing their understanding of course material. • Led small group discussions, review sessions, and problem-solving activities to reinforce course concepts. • Graded assignments, quizzes, and exams while providing constructive feedback to students. • Assisted in developing course materials and maintained consistency in course delivery. • Participated in professional development activities to enhance teaching skills. • Held regular office hours and utilized instructional technologies to support student learning.
05/2015 – 05/2016	Teaching and Research Assistant, KNUST	Ghana	• Organized tutorials and graded students' work in ocular epidemiology, serving as a liaison for junior statistics students' programs at KNUST. • Engaged students in remedial and extra-class tutorials to prepare them for academic, personal, and professional success. • Graded assignments, quizzes, and exams while providing constructive feedback to students.

EDUCATION

08/2020 – Present	University of Alabama at Birmingham (UAB), School of Public Health PhD, Biostatistics Dissertation Title: Statistical methods for Mendelian Randomization Studies.	Birmingham - Alabama
08/2016 – 05/2018	Northern Arizona University MSc., Statistics	Arizona
08/2011 – 05/2015	Kwame Nkrumah University of Science and Technology BSc, Statistics First Class Hons.	Kumasi, GHANA

SKILLS

R	SAS
SPSS	JMP
multivariate regression analysis	Time series analysis
Survival analysis	Bayesian analysis
Analytical skills	Scientific writing
Communication	Presentation skills

CERTIFICATES

Bayesian Thinking in Clinical Research – Statement of Completion UCSF-Stanford CERSI	2025
UAB center for Teaching and Learning Certificate UAB	2024
SAS Certified Specialist: Base Programming Using SAS 9.4 SAS	2021
Introduction to Probability and Data with R Duke University (Coursera)	2017